

Fundamental Calculus/Differential and Integral Calculus

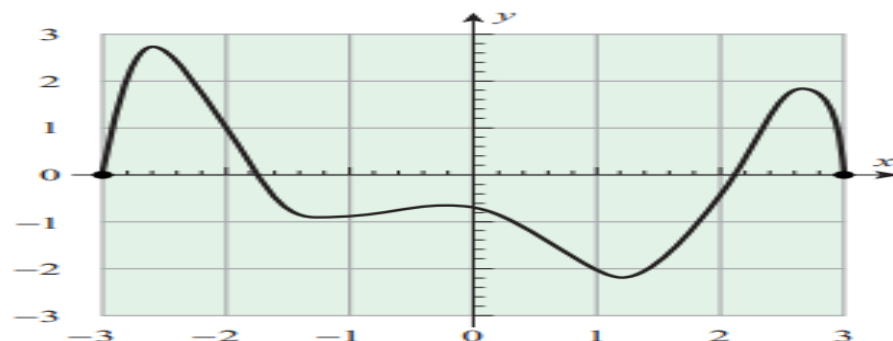
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Reference : Calculus 10-th Edition by Howard Anton,
Irl Bivens and Stephen Davis

Function: Domain and Range

- Section: 0.1

1. Use the accompanying graph to answer the following questions, making reasonable approximations where needed.
- (a) For what values of x is $y = 1$?
 - (b) For what values of x is $y = 3$?
 - (c) For what values of y is $x = 3$?
 - (d) For what values of x is $y \leq 0$?
 - (e) What are the maximum and minimum values of y and for what values of x do they occur?



◀ Figure Ex-1

1. (a) $-2.9, -2.0, 2.35, 2.9$ (b) None (c) $y = 0$ (d) $-1.75 \leq x \leq 2.15, x = -3, x = 3$
- (e) $y_{\max} = 2.8$ at $x = -2.6$; $y_{\min} = -2.2$ at $x = 1.2$

Function

7. Find $f(0)$, $f(2)$, $f(-2)$, $f(3)$, $f(\sqrt{2})$, and $f(3t)$.

$$(a) \ f(x) = 3x^2 - 2 \qquad (b) \ f(x) = \begin{cases} \frac{1}{x}, & x > 3 \\ 2x, & x \leq 3 \end{cases}$$

8. Find $g(3)$, $g(-1)$, $g(\pi)$, $g(-1.1)$, and $g(t^2 - 1)$.

$$(a) \ g(x) = \frac{x+1}{x-1} \qquad (b) \ g(x) = \begin{cases} \sqrt{x+1}, & x \geq 1 \\ 3, & x < 1 \end{cases}$$

Function

7. (a) $f(0) = 3(0)^2 - 2 = -2$; $f(2) = 3(2)^2 - 2 = 10$; $f(-2) = 3(-2)^2 - 2 = 10$; $f(3) = 3(3)^2 - 2 = 25$; $f(\sqrt{2}) = 3(\sqrt{2})^2 - 2 = 4$; $f(3t) = 3(3t)^2 - 2 = 27t^2 - 2$.

(b) $f(0) = 2(0) = 0$; $f(2) = 2(2) = 4$; $f(-2) = 2(-2) = -4$; $f(3) = 2(3) = 6$; $f(\sqrt{2}) = 2\sqrt{2}$; $f(3t) = 1/(3t)$ for $t > 1$ and $f(3t) = 6t$ for $t \leq 1$.

8. (a) $g(3) = \frac{3+1}{3-1} = 2$; $g(-1) = \frac{-1+1}{-1-1} = 0$; $g(\pi) = \frac{\pi+1}{\pi-1}$; $g(-1.1) = \frac{-1.1+1}{-1.1-1} = \frac{-0.1}{-2.1} = \frac{1}{21}$; $g(t^2 - 1) = \frac{t^2 - 1 + 1}{t^2 - 1 - 1} = \frac{t^2}{t^2 - 2}$.

(b) $g(3) = \sqrt{3+1} = 2$; $g(-1) = 3$; $g(\pi) = \sqrt{\pi+1}$; $g(-1.1) = 3$; $g(t^2 - 1) = 3$ if $t^2 < 2$ and $g(t^2 - 1) = \sqrt{t^2 - 1 + 1} = |t|$ if $t^2 \geq 2$.

Finding Domain and range of

$$y = f(x) = \frac{p(x)}{q(x)} \quad [q(x) \neq 0] \text{ (Rational function)}$$

- 1. we have to check whether there is any value of x for which y is undefined from $y = f(x) = \frac{p(x)}{q(x)} \quad [q(x) \neq 0]$.
- 2. we have to check whether there is any value of y for which x is undefined from
- $x = f(y) = \frac{p(x)}{q(x)} \quad [q(x) \neq 0]$.
- 3. we have to ignore undefined, complex value for the given function.

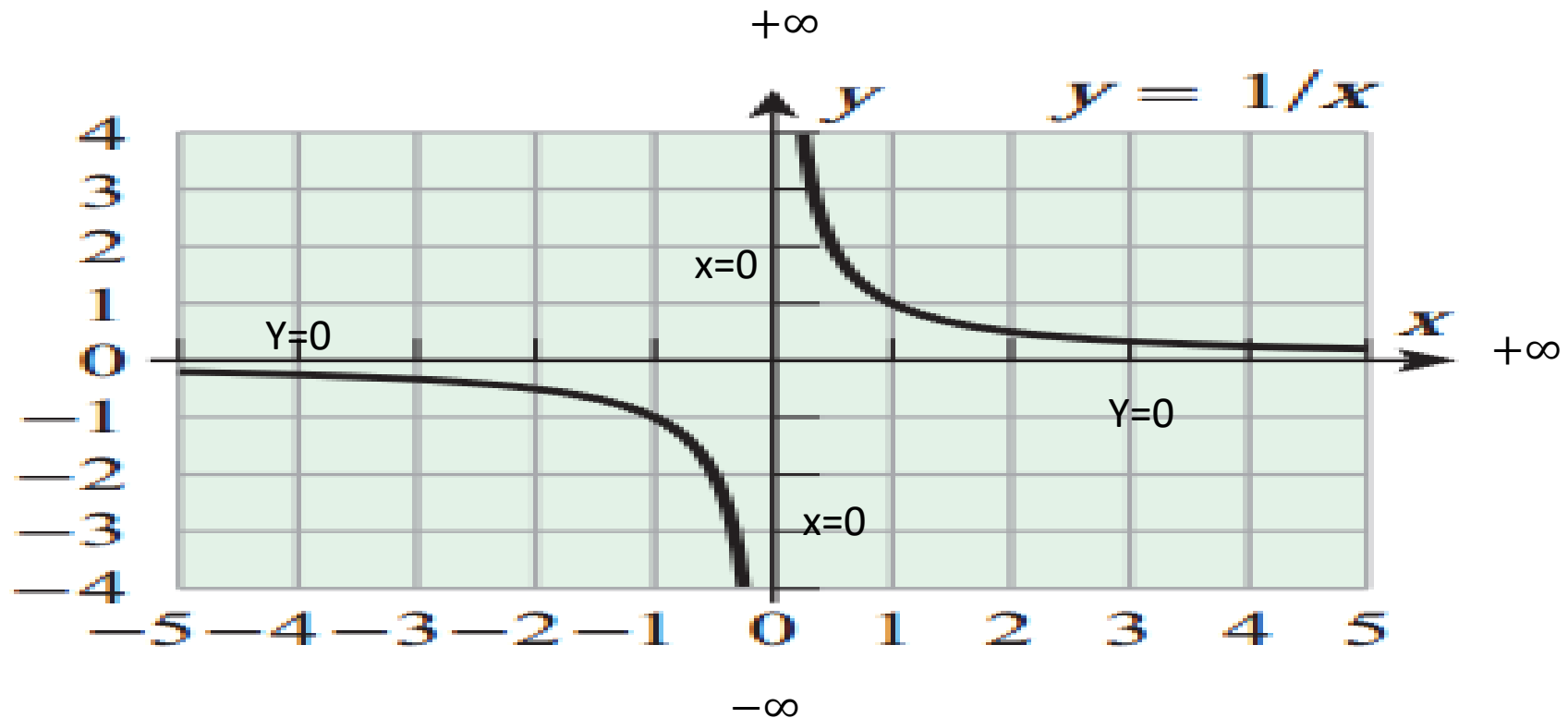
Finding Domain and range of $y = f(x) = \frac{1}{x}, x \neq 0$.

(Rational function $\frac{p(x)}{q(x)}$ form)

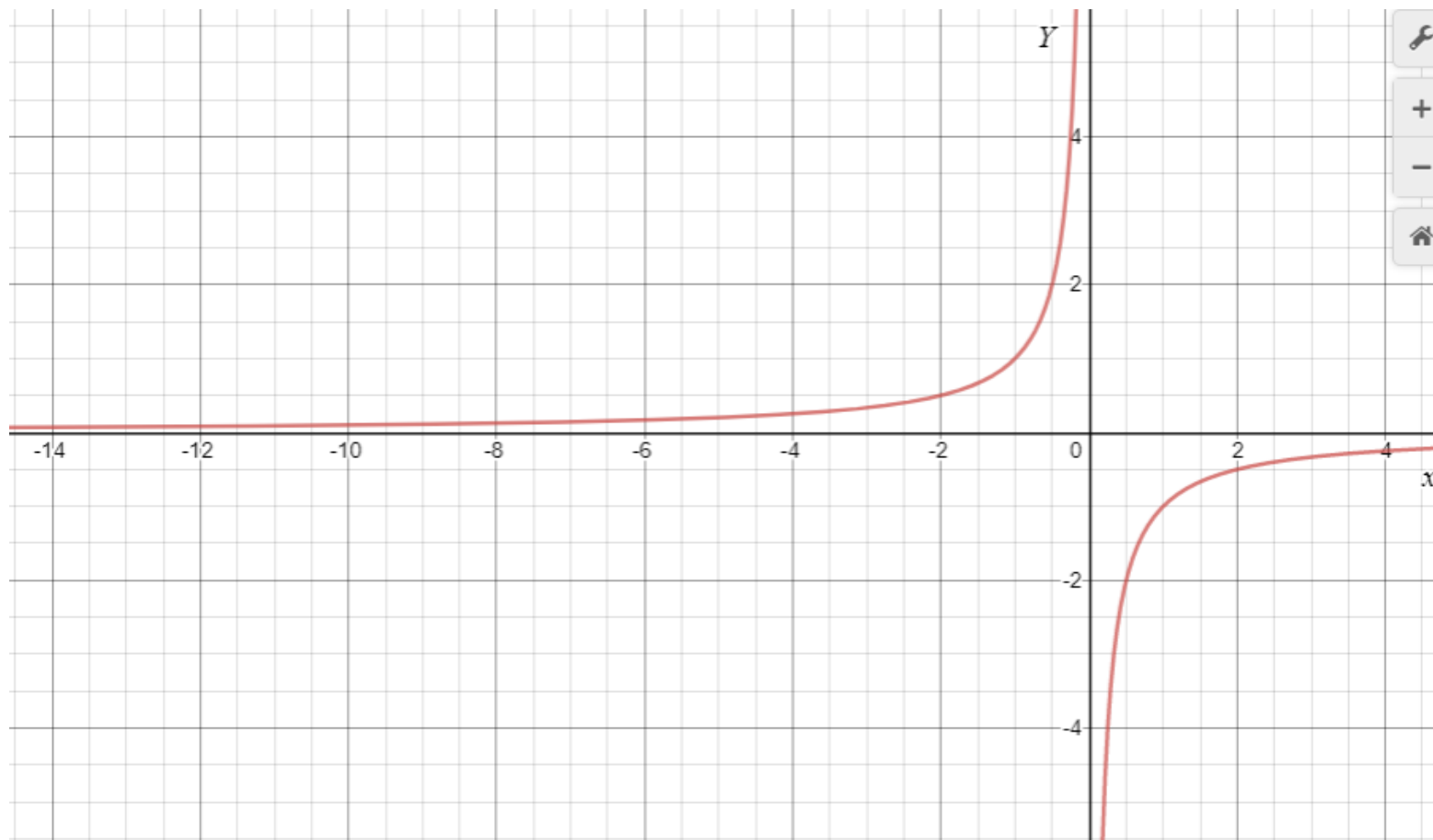
Since f is undefined if $x = 0$, so domain is $x \neq 0$.

$$y = \frac{1}{x} \Rightarrow x = \frac{1}{y}, y \neq 0$$

and range is $y \neq 0$



Graph of $f(x)=y = -\frac{1}{x}$



Show all

×

Sketch the graph of $f(x)=y = 1 + \frac{1}{x}$, also find their domain and range.

- Since f is undefined if $x = 0$, so domain is $x \neq 0$.
- Here $y = 1 + \frac{1}{x} \Rightarrow y - 1 = \frac{1}{x} \Rightarrow xy - x = 1$
- $\Rightarrow x(y - 1) = 1 \Rightarrow x = \frac{1}{(y-1)}, y \neq 1$

, so range is $y \neq 1$

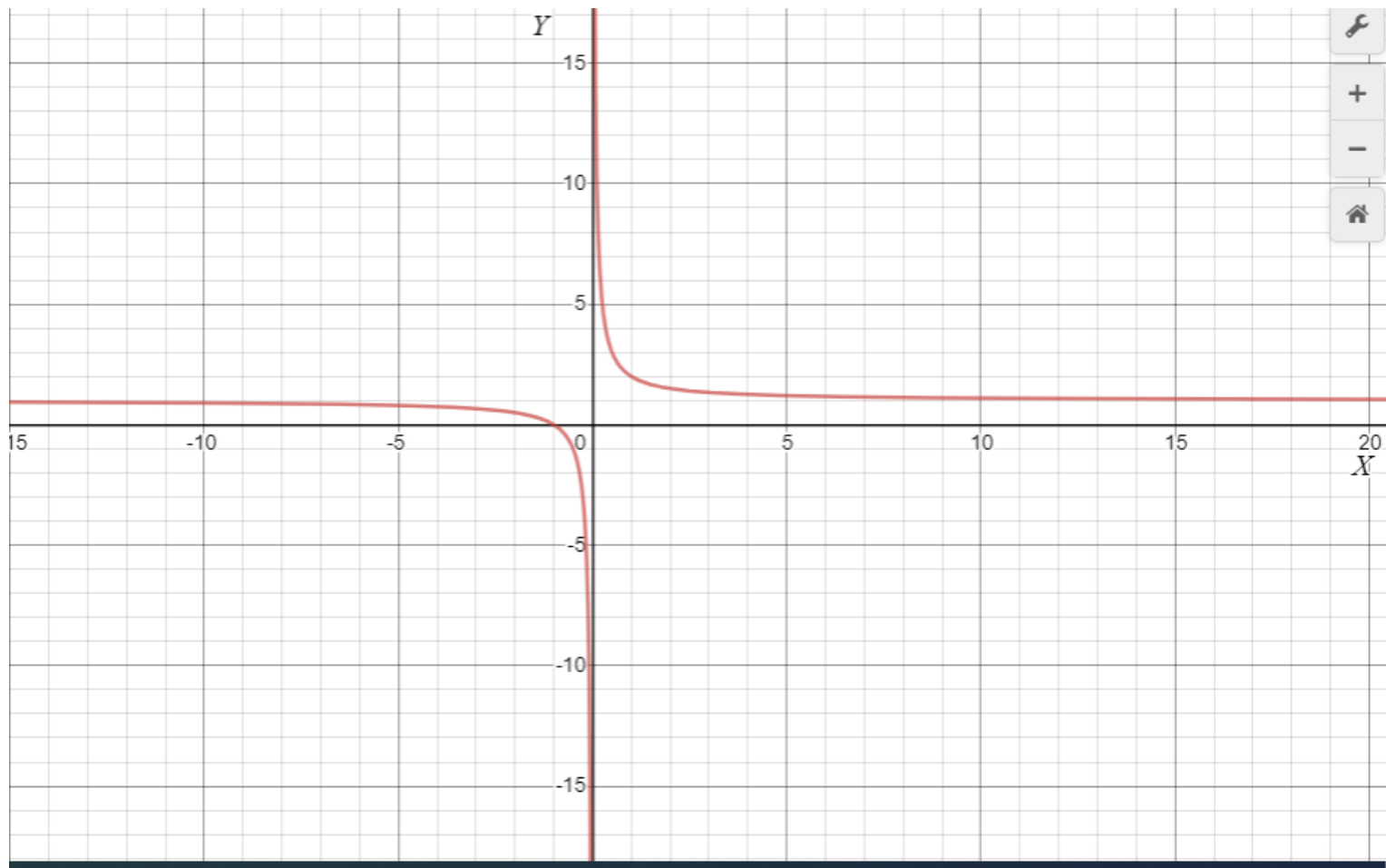
Asymptote line $x = 0, y = 0 + 1 = 1$,

Or

Domain is : $R - \{0\}$ and Range is : $R - \{1\}$ or ,

Domain is $(-\infty, 0) \cup (0, +\infty)$ and Range is $(-\infty, 1) \cup (1, +\infty)$

Graph of $f(x)=y = 1 + \frac{1}{x}$



Sketch the graph of $f(x)=y = 1 - \frac{1}{x}$, also find their domain and range.

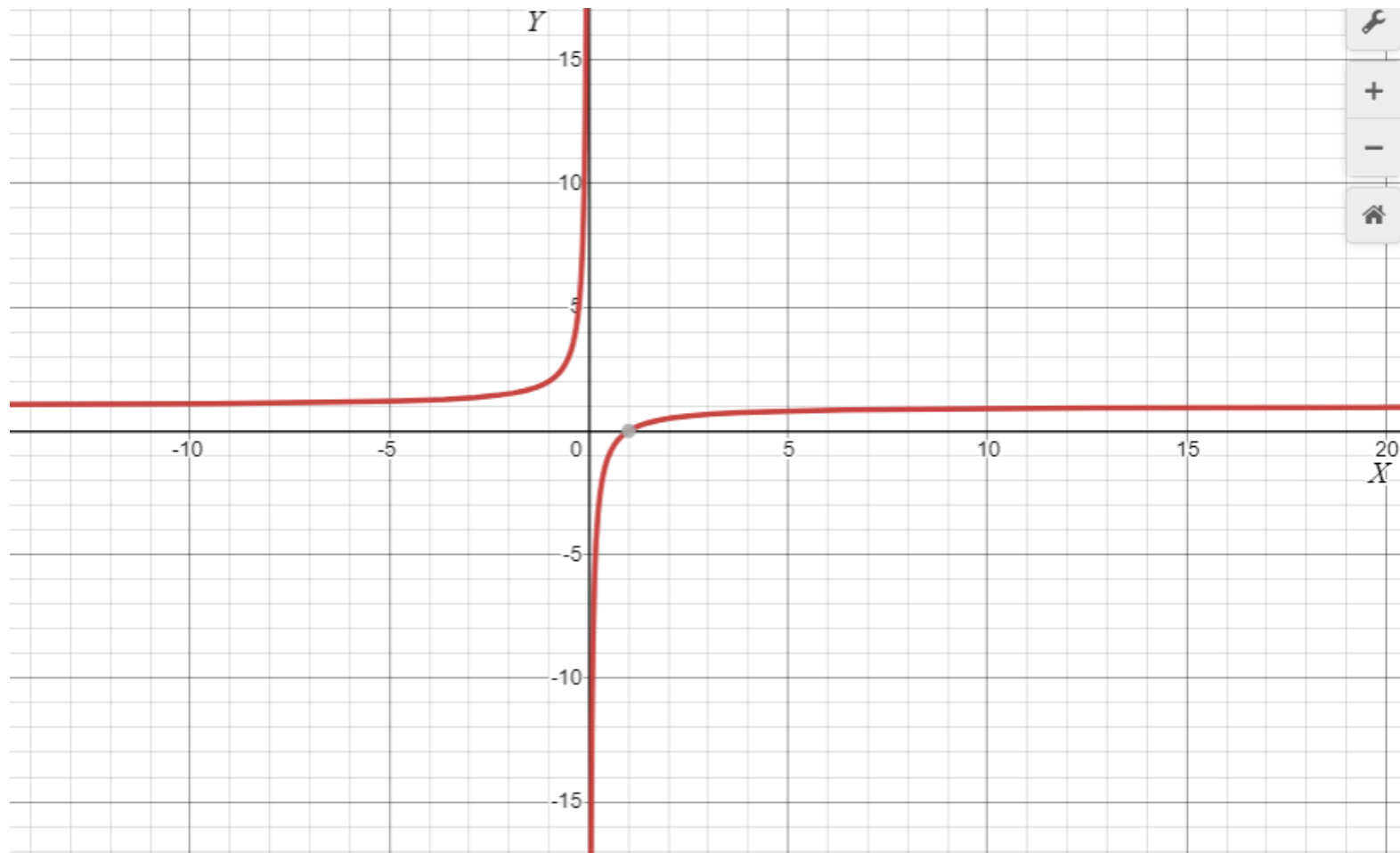
- Since f is undefined if $x = 0$, so domain is $x \neq 0$.
- Here $y = 1 - \frac{1}{x} \Rightarrow -y + 1 = \frac{1}{x} \Rightarrow -xy + x = 1$
- $\Rightarrow x(1 - y) = 1 \Rightarrow x = \frac{1}{1-y}, y \neq 1$
- , , so range is $y \neq 1$

Or

Domain is : $R - \{0\}$ and Range is : $R - \{1\}$ or ,

Domain is $(-\infty, 0) \cup (0, +\infty)$ and Range is $(-\infty, 1) \cup (1, +\infty)$

Graph of $f(x)=y = 1 - \frac{1}{x}$

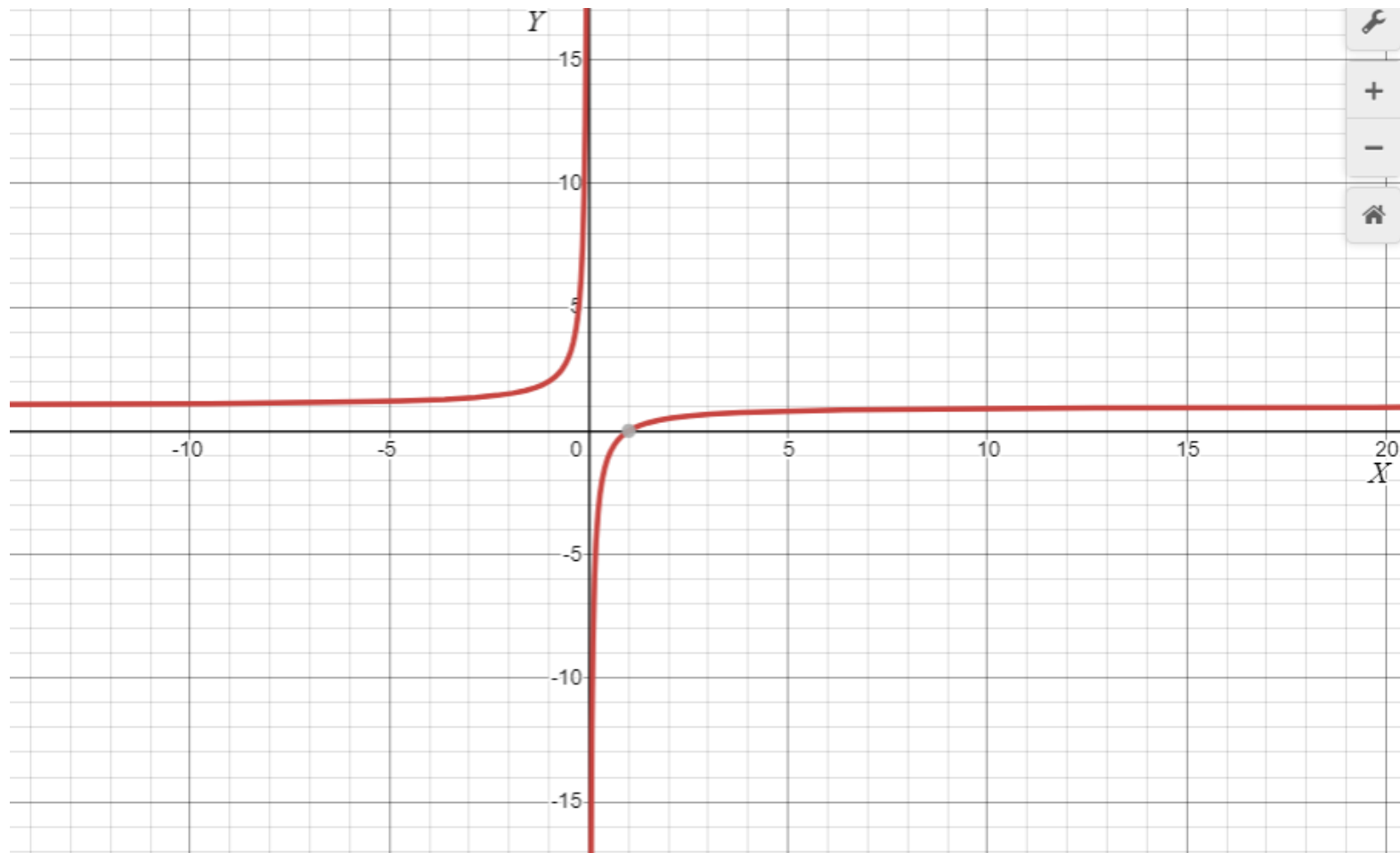


Home work

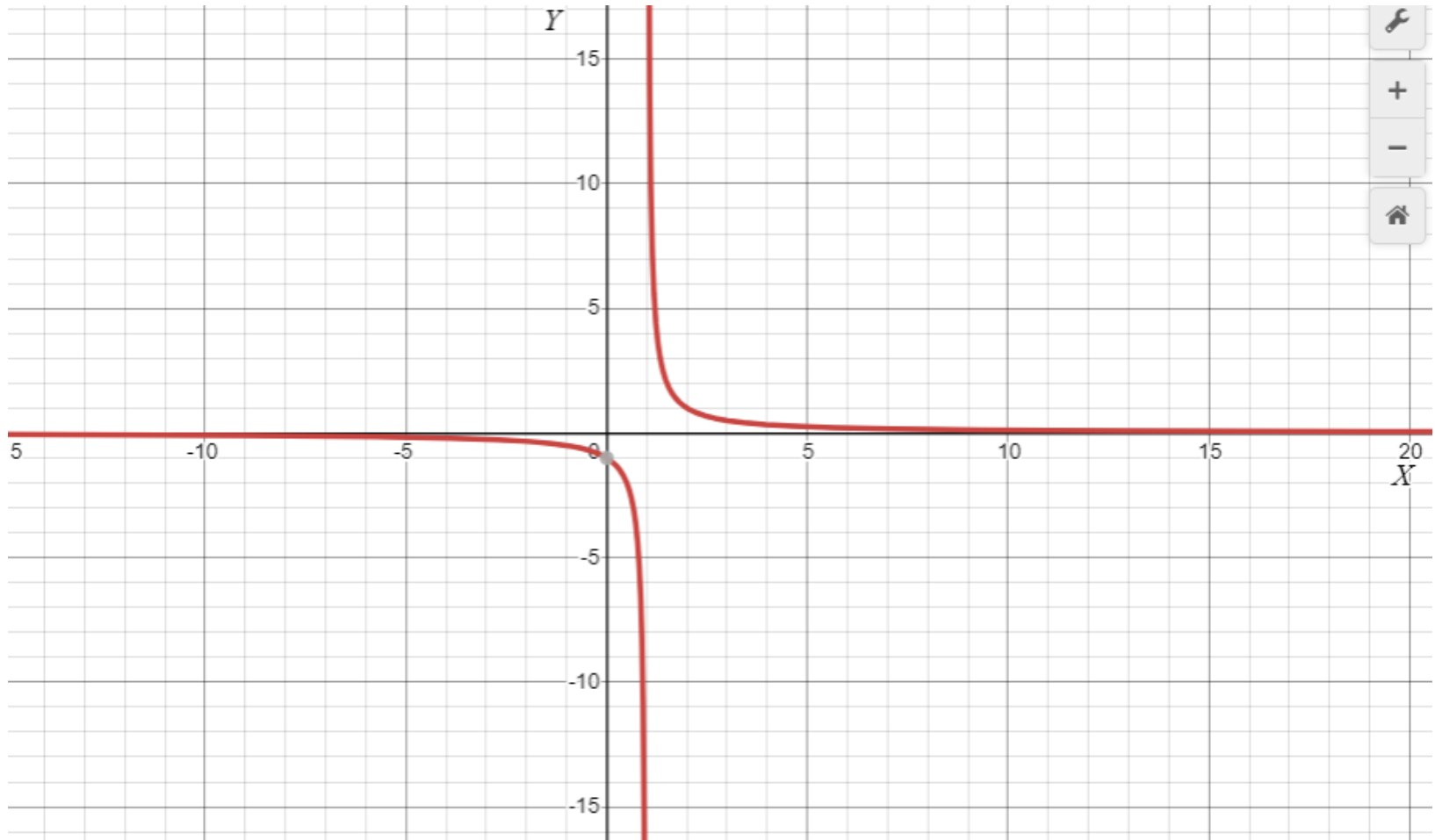
- $y = \frac{x-1}{x} = \frac{x}{x} - \frac{1}{x} = 1 - \frac{1}{x}$ [Home work]

- $y = \frac{x}{x-1} = \frac{x-1+1}{x-1} = \frac{x-1}{x-1} + \frac{1}{x-1}$
 $= 1 + \frac{1}{x-1}$ [Home work] $x=1, y=1$

Graph of $f(x)=y = 1 - \frac{1}{x}$



Sketch the graph of $f(x)=y = \frac{1}{x-1}$, also find their domain and range.

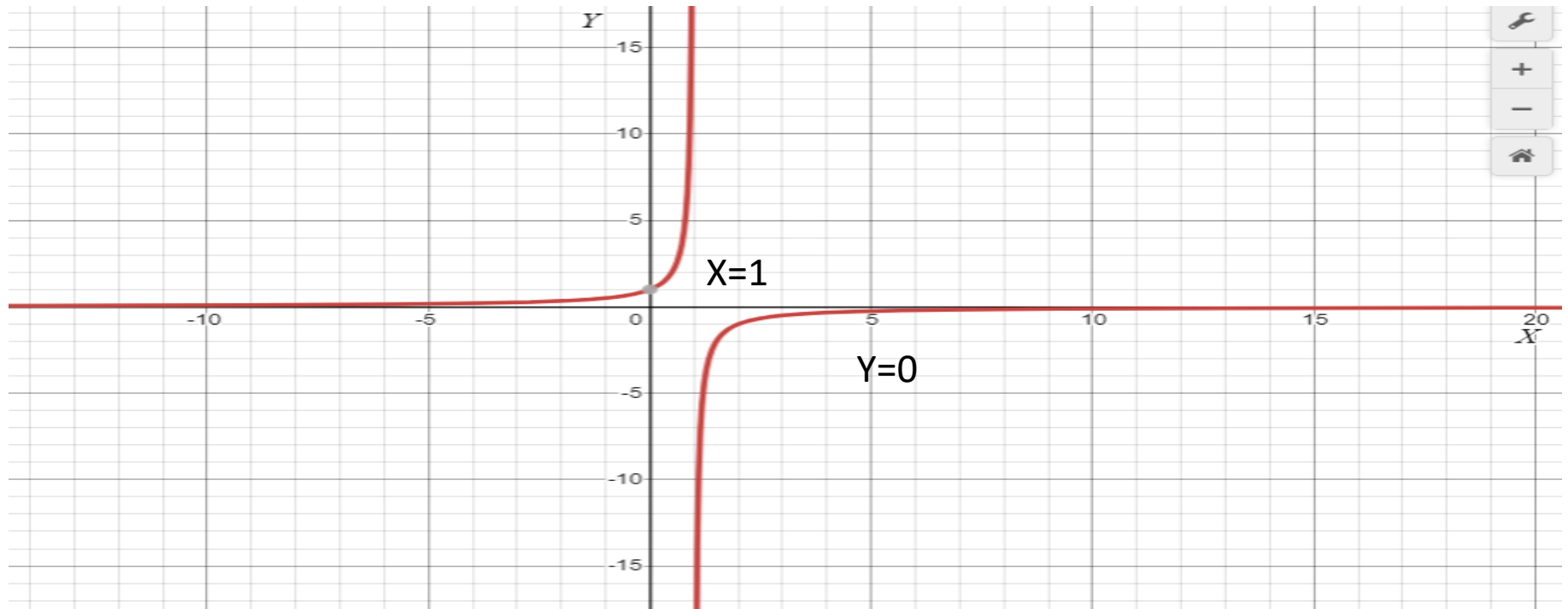


Finding domain and range $y = \frac{1}{x-1}$.

- Since f is undefined if $x - 1 = 0$
 $\Rightarrow x = 1$ so domain is $x \neq 1$.
- Here $y = \frac{1}{x-1} \Rightarrow xy - y = 1$
- $\Rightarrow xy = 1 - y \Rightarrow x = \frac{1+y}{y}, y \neq 0$.
- So range is $y \neq 0$
- Or
- Domain is : $R - \{1\}$ and Range is : $R - \{0\}$

Sketch the graph of $f(x)=y = \frac{1}{1-x}$,
also find their domain and range.

- $y = \frac{1}{1-x} = \frac{1}{-(x-1)}$



Finding domain and range $y = \frac{1}{1-x}$.

- Since f is undefined if $1-x = 0$
 $\Rightarrow x = 1$ so domain is $x \neq 1$.
- Here $y = \frac{1}{1-x} \Rightarrow -xy + y = 1$
- $\Rightarrow xy = 1 - y \Rightarrow x = \frac{1-y}{y}, y \neq 0$.
- So range is $y \neq 0$
- Or
- Domain is : $R - \{1\}$ and Range is : $R - \{0\}$

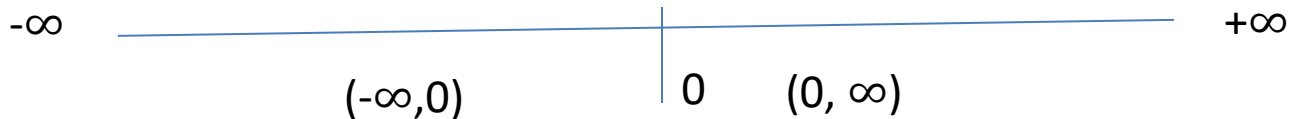
Finding domain and range $y = \frac{1}{1-x}$.

- Since f is undefined if $1-x = 0$
 $\Rightarrow x = 1$ so domain is $x \neq 1$.
- Here $y = \frac{1}{1-x} \Rightarrow -xy + y = 1$
- $\Rightarrow xy = 1 - y \Rightarrow x = \frac{1-y}{y}, y \neq 0$.
- So range is $y \neq 0$
- Or
- Domain is : $R - \{1\}$ and Range is : $R - \{0\}$

or

- Domain is : $R - \{0\}$ and
- Range is : $R - \{0\}$

Or



- Domain is $(-\infty, 0) \cup (0, +\infty)$
- Range is $(-\infty, 0) \cup (0, +\infty)$

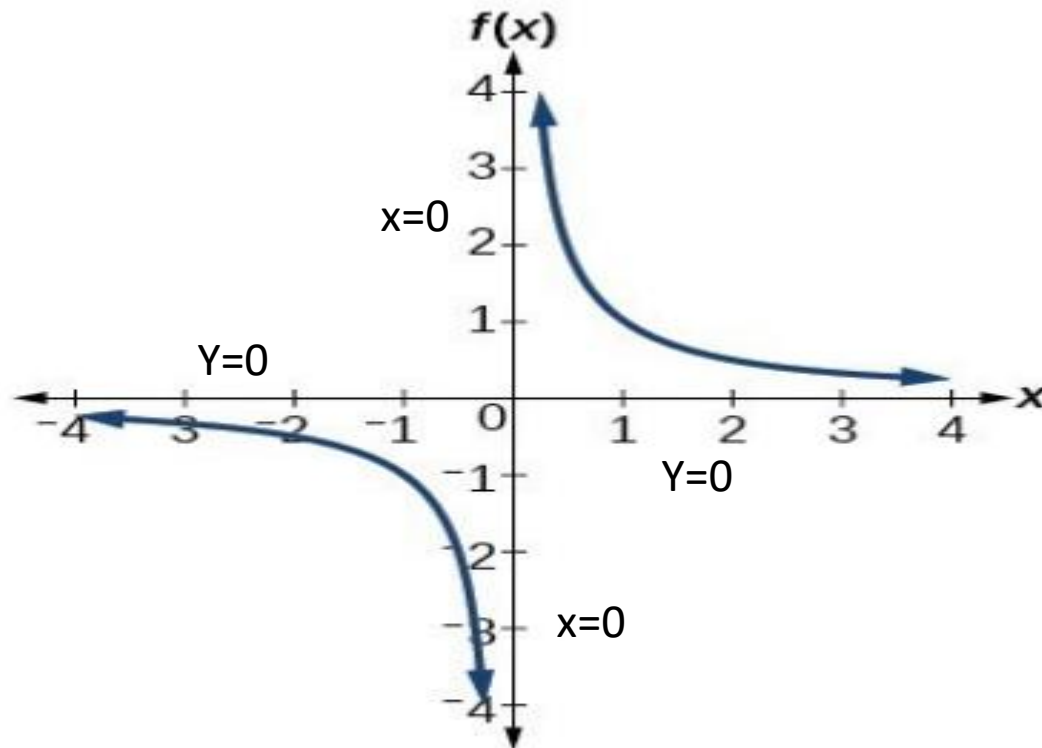
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Input	Output
$x = -\infty$	$y = 0$
$x = -1$	$y = -1$
$x = -\frac{1}{2}$	$y = -2$
$x = -\frac{1}{3}$	$y = -3$
$x = 0$	$y = -\infty$ $y = +\infty$
$x = 1$	$y = 1$
$x = \frac{1}{2}$	$y = -2$
$x = -\frac{1}{3}$	$y = -3$
$x = +\infty$	$y = 0$

Asymptote

- An asymptote is a line that the graph of function approaches but never touches. Rational function contain asymptote.
- $x = -\infty, y = 0, x = \infty, y = 0$
- $x = 0, y = -\infty, x = 0, y = +\infty$
- so here asymptote line are $x = 0$ and $y = 0$
 $x = 0$, asymptote(Vertical),
so domain is $x \neq 0$.
 $y = 0$, asymptote(Horizontal),
so range is $y \neq 0$

Graph of function $y = f(x) = \frac{1}{x}$



Domain: $(-\infty, 0) \cup (0, \infty)$

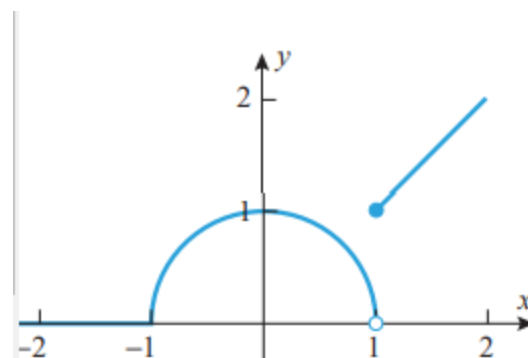
Range: $(-\infty, 0) \cup (0, \infty)$

■ PIECEWISE-DEFINED FUNCTIONS

The absolute value function $f(x) = |x|$ is an example of a function that is defined *piecewise* in the sense that the formula for f changes, depending on the value of x .

► **Example 4** Sketch the graph of the function defined piecewise by the formula

$$f(x) = \begin{cases} 0, & x \leq -1 \\ \sqrt{1-x^2}, & -1 < x < 1 \\ x, & x \geq 1 \end{cases}$$



▲ Figure 0.1.10

Natural domain

► **Example 6** Find the natural domain of

(a) $f(x) = x^3$ (b) $f(x) = 1/[(x - 1)(x - 3)]$

(d) $f(x) = \sqrt{x^2 - 5x + 6}$

Solution (a). The function f has real values for all real x , so its natural domain is the interval $(-\infty, +\infty)$.

Solution (b). The function f has real values for all real x , except $x = 1$ and $x = 3$, where divisions by zero occur. Thus, the natural domain is

$$\{x : x \neq 1 \text{ and } x \neq 3\} = (-\infty, 1) \cup (1, 3) \cup (3, +\infty)$$

Solution (d). The function f has real values, except when the expression inside the radical is negative. Thus the natural domain consists of all real numbers x such that

$$x^2 - 5x + 6 = (x - 3)(x - 2) \geq 0$$

This inequality is satisfied if $x \leq 2$ or $x \geq 3$ (verify), so the natural domain of f is

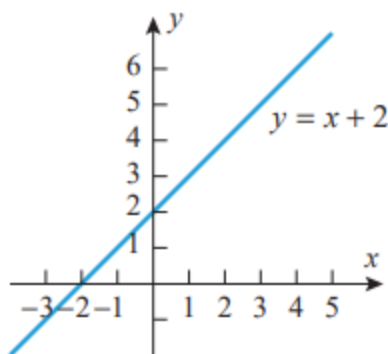
$$(-\infty, 2] \cup [3, +\infty) \quad \blacktriangleleft$$

► **Example 7** The natural domain of the function

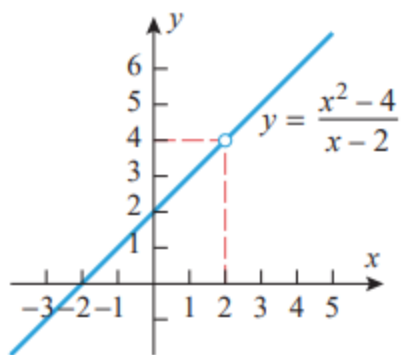
$$f(x) = \frac{x^2 - 4}{x - 2} \quad (2)$$

consists of all real x except $x = 2$. However, if we factor the numerator and then cancel the common factor in the numerator and denominator, we obtain

$$f(x) = \frac{(x - 2)(x + 2)}{x - 2} = x + 2 \quad (3)$$



(a)

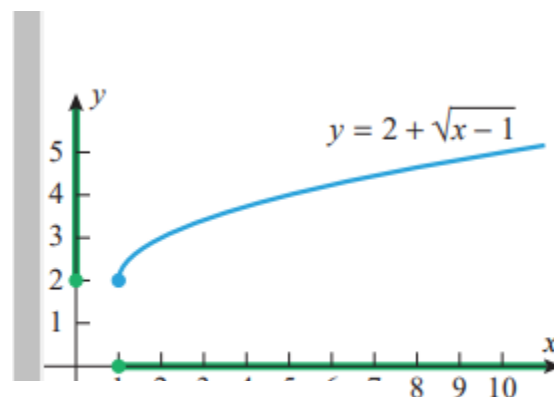


(b)

$$f(x) = x + 2, \quad x \neq 2$$

► **Example 8** Find the domain and range of

(a) $f(x) = 2 + \sqrt{x - 1}$



Solution (a). Since no domain is stated explicitly, the domain of f is its natural domain, $[1, +\infty)$. As x varies over the interval $[1, +\infty)$, the value of $\sqrt{x - 1}$ varies over the interval $[0, +\infty)$, so the value of $f(x) = 2 + \sqrt{x - 1}$ varies over the interval $[2, +\infty)$, which is the range of f . The domain and range are highlighted in green on the x - and y -axes in Figure 0.1.15.

(b) $f(x) = (x + 1)/(x - 1)$

To determine the range it will be convenient to introduce a dependent variable

$$y = \frac{x + 1}{x - 1} \quad (4)$$

To see that this is so, we solve (4) for x in terms of y :

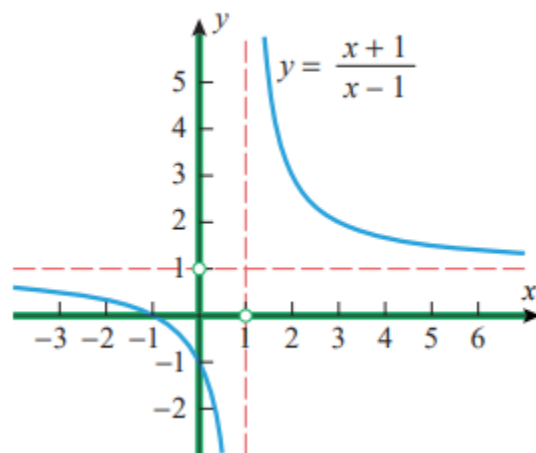
$$(x - 1)y = x + 1$$

$$xy - y = x + 1$$

$$xy - x = y + 1$$

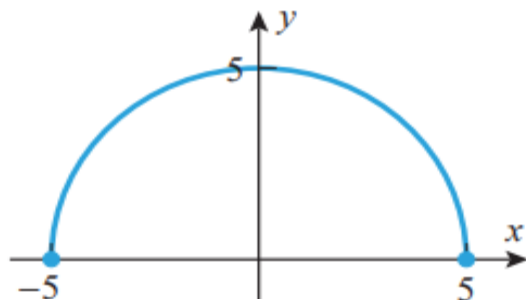
$$x(y - 1) = y + 1$$

$$x = \frac{y + 1}{y - 1}$$



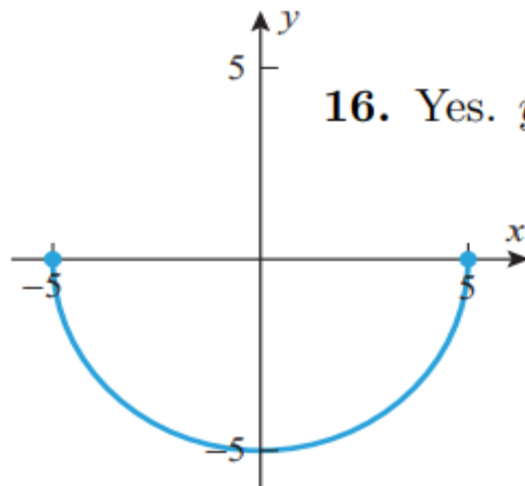
15–18 As seen in Example 3, the equation $x^2 + y^2 = 25$ does not define y as a function of x . Each graph in these exercises is a portion of the circle $x^2 + y^2 = 25$. In each case, determine whether the graph defines y as a function of x , and if so, give a formula for y in terms of x . ■

15.



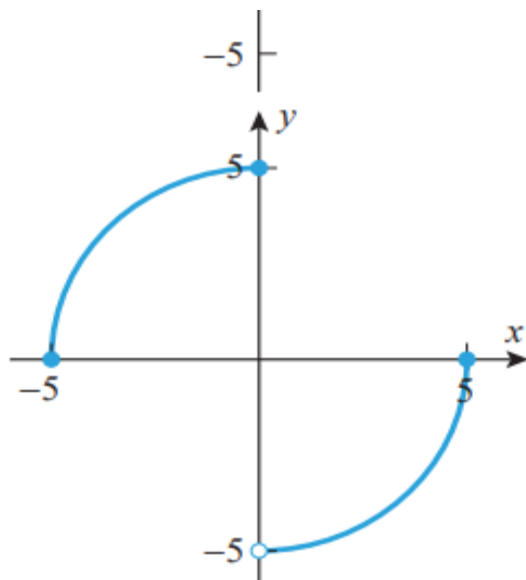
15. Yes. $y = \sqrt{25 - x^2}$.

16.

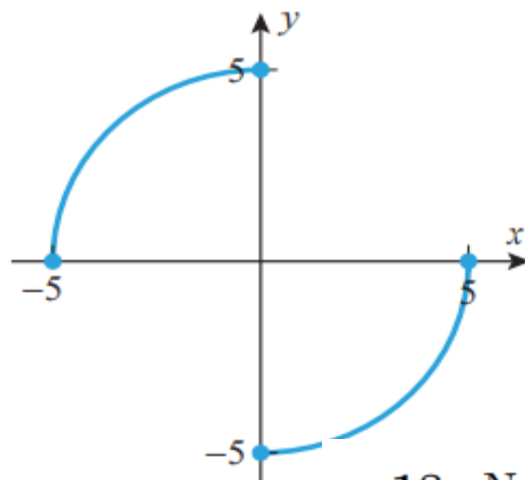


16. Yes. $y = -\sqrt{25 - x^2}$.

17.



18.



18. No; the vertical line $x =$

- 9–10 Find the natural domain and determine the range of each function. If you have a graphing utility, use it to confirm that your result is consistent with the graph produced by your graphing utility.

$$9. \text{ (a) } f(x) = \frac{1}{x-3}$$

$$\text{(c) } g(x) = \sqrt{x^2 - 3}$$

$$\text{(b) } F(x) = \frac{x}{|x|}$$

$$\text{(d) } G(x) = \sqrt{x^2 - 2x + 5}$$

$$\text{(f) } H(x) = \sqrt{\frac{x^2 - 4}{x - 2}}$$

$$\text{(a) } f(x) = \sqrt{3 - x}$$

$$\text{(c) } g(x) = 3 + \sqrt{x}$$

$$\text{(b) } F(x) = \sqrt{4 - x^2}$$

$$\text{(d) } G(x) = x^3 + 2$$

Chapter : 0.2(Page-15)

NEW FUNCTIONS FROM OLD

NEW FUNCTIONS FROM OLD

■ ARITHMETIC OPERATIONS ON FUNCTIONS

Two functions, f and g , can be added, subtracted, multiplied, and divided in a natural way to form new functions $f + g$, $f - g$, fg , and f/g . For example, $f + g$ is defined by the formula

$$(f + g)(x) = f(x) + g(x) \tag{1}$$

0.2.1 DEFINITION Given functions f and g , we define

$$(f + g)(x) = f(x) + g(x)$$

$$(f - g)(x) = f(x) - g(x)$$

$$(fg)(x) = f(x)g(x)$$

$$(f/g)(x) = f(x)/g(x)$$

For the functions $f + g$, $f - g$, and fg we define the domain to be the intersection of the domains of f and g , and for the function f/g we define the domain to be the intersection of the domains of f and g but with the points where $g(x) = 0$ excluded (to avoid division by zero).

► **Example 1** Let

$$f(x) = 1 + \sqrt{x-2} \quad \text{and} \quad g(x) = x - 3$$

Find the domains and formulas for the functions $f + g$, $f - g$, fg , f/g , and $7f$.

Solution. First, we will find the formulas and then the domains. The formulas are

$$(f + g)(x) = f(x) + g(x) = (1 + \sqrt{x-2}) + (x - 3) = x - 2 + \sqrt{x-2} \quad (2)$$

$$(f - g)(x) = f(x) - g(x) = (1 + \sqrt{x-2}) - (x - 3) = 4 - x + \sqrt{x-2} \quad (3)$$

$$(fg)(x) = f(x)g(x) = (1 + \sqrt{x-2})(x - 3) \quad (4)$$

$$(f/g)(x) = f(x)/g(x) = \frac{1 + \sqrt{x-2}}{x - 3} \quad (5)$$

$$(7f)(x) = 7f(x) = 7 + 7\sqrt{x-2} \quad (6)$$

The domains of f and g are $[2, +\infty)$ and $(-\infty, +\infty)$, respectively (their natural domains). Thus, it follows from Definition 0.2.1 that the domains of $f + g$, $f - g$, and fg are the intersection of these two domains, namely,

$$[2, +\infty) \cap (-\infty, +\infty) = [2, +\infty) \quad (7)$$

► **Example 2** Show that if $f(x) = \sqrt{x}$, $g(x) = \sqrt{x}$, and $h(x) = x$, then the domain of fg is not the same as the natural domain of h .

Solution. The natural domain of $h(x) = x$ is $(-\infty, +\infty)$. Note that

$$(fg)(x) = \sqrt{x}\sqrt{x} = x = h(x)$$

on the domain of fg . The domains of both f and g are $[0, +\infty)$, so the domain of fg is

$$[0, +\infty) \cap [0, +\infty) = [0, +\infty)$$

■ COMPOSITION OF FUNCTIONS

0.2.2 DEFINITION Given functions f and g , the *composition* of f with g , denoted by $f \circ g$, is the function defined by

$$(f \circ g)(x) = f(g(x))$$

The domain of $f \circ g$ is defined to consist of all x in the domain of g for which $g(x)$ is in the domain of f .

suppose that

$$f(x) = x^2 \quad \text{and} \quad g(x) = x + 1$$

If we substitute $g(x)$ for x in the formula for f , we obtain a new function

$$f(g(x)) = (g(x))^2 = (x + 1)^2$$

which we denote by $f \circ g$. Thus,

$$(f \circ g)(x) = f(g(x)) = (g(x))^2 = (x + 1)^2$$

► **Example 3** Let $f(x) = x^2 + 3$ and $g(x) = \sqrt{x}$. Find

(a) $(f \circ g)(x)$ (b) $(g \circ f)(x)$

Solution (a). The formula for $f(g(x))$ is

$$f(g(x)) = [g(x)]^2 + 3 = (\sqrt{x})^2 + 3 = x + 3$$

Since the domain of g is $[0, +\infty)$ and the domain of f is $(-\infty, +\infty)$, the domain of $f \circ g$ consists of all x in $[0, +\infty)$ such that $g(x) = \sqrt{x}$ lies in $(-\infty, +\infty)$; thus, the domain of $f \circ g$ is $[0, +\infty)$. Therefore,

$$(f \circ g)(x) = x + 3, \quad x \geq 0$$

Solution (b). The formula for $g(f(x))$ is

$$g(f(x)) = \sqrt{f(x)} = \sqrt{x^2 + 3}$$

Since the domain of f is $(-\infty, +\infty)$ and the domain of g is $[0, +\infty)$, the domain of $g \circ f$ consists of all x in $(-\infty, +\infty)$ such that $f(x) = x^2 + 3$ lies in $[0, +\infty)$. Thus, the domain of $g \circ f$ is $(-\infty, +\infty)$. Therefore,

$$(g \circ f)(x) = \sqrt{x^2 + 3}$$

There is no need to indicate that the domain is $(-\infty, +\infty)$, since this is the natural domain of $\sqrt{x^2 + 3}$. ◀

► **Example 4** Find $(f \circ g \circ h)(x)$ if

$$f(x) = \sqrt{x}, \quad g(x) = 1/x, \quad h(x) = x^3$$

Solution.

$$(f \circ g \circ h)(x) = f(g(h(x))) = f(g(x^3)) = f(1/x^3) = \sqrt{1/x^3} = 1/x^{3/2} \quad \blacktriangleleft$$

► **Example 5** Express $\sin(x^3)$ as a composition of two functions.

Solution. To evaluate $\sin(x^3)$, we would first compute x^3 and then take the sine, so $g(x) = x^3$ is the inside function and $f(x) = \sin x$ the outside function. Therefore,

$$\sin(x^3) = f(g(x)) \quad \boxed{g(x) = x^3 \text{ and } f(x) = \sin x} \quad \blacktriangleleft$$

Intercepts

- If we put $x = 0$, we get y intercepts, if we put
- $y = 0$, we get x intercepts.

Find the intercepts of $f(x) = x^2 - 4x$. If we put $y = 0$, $y = f(x) = x^2 - 4x = 0$, if $y = 0$

$$x(x - 4) = 0$$

$\Rightarrow x = 0, x = 4$, so x intercepts are $(x, y) = (0, 0)$ and $(4, 0)$, If we put $x = 0$, $y = f(0) = 0^2 - 4 \cdot 0 = 0$, so y intercepts are $(x, y) = (0, 0)$

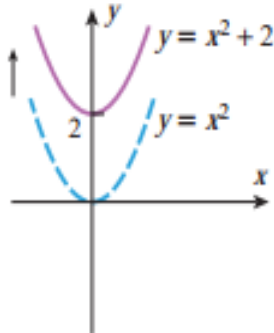
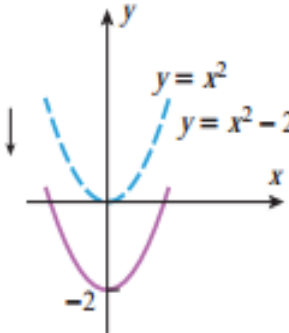
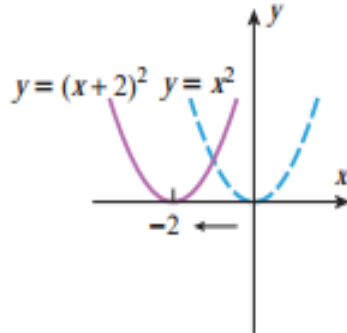
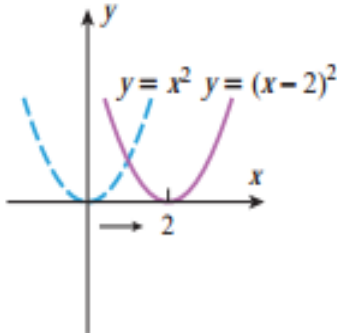
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0.2

NEW FUNCTIONS FROM OLD

TRANSLATIONS (Page-20) :**Table 0.2.2**

TRANSLATION PRINCIPLES

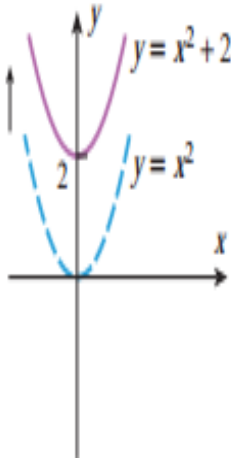
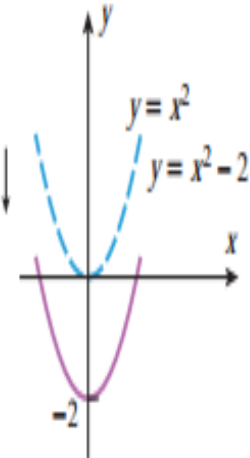
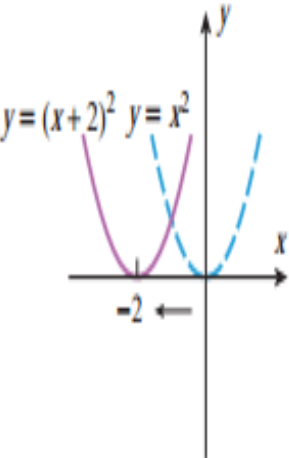
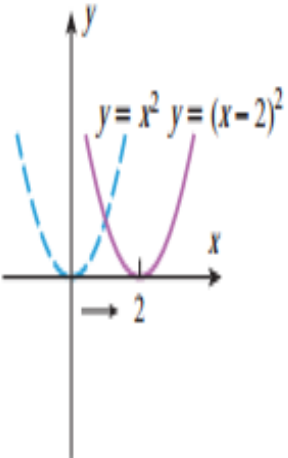
OPERATION ON $y = f(x)$	Add a positive constant c to $f(x)$	Subtract a positive constant c from $f(x)$	Add a positive constant c to x	Subtract a positive constant c from x
NEW EQUATION	$y = f(x) + c$	$y = f(x) - c$	$y = f(x + c)$	$y = f(x - c)$
GEOMETRIC EFFECT	Translates the graph of $y = f(x)$ up c units	Translates the graph of $y = f(x)$ down c units	Translates the graph of $y = f(x)$ left c units	Translates the graph of $y = f(x)$ right c units
EXAMPLE				

NEW FUNCTIONS FROM OLD

- Table 0.2.2 illustrates the geometric effect on the graph of $y = f(x)$ of adding or subtracting a *positive* constant c to f or to its independent variable x . For example, the first result in the table illustrates that adding a positive constant c to a function f adds c to each y -coordinate of its graph, thereby shifting the graph of f up by c units.
- Similarly, subtracting c from f shifts the graph down by c units. On the other hand, if a positive constant c is added to x , then the value of $y = f(x + c)$ at $x - c$ is $f(x)$; and since the point $x - c$ is c units to the left of x on the x -axis, the graph of $y = f(x + c)$ must be the graph of $y = f(x)$ shifted left by c units.
- Similarly, subtracting c from x shifts the graph of $y = f(x)$ right by c units.

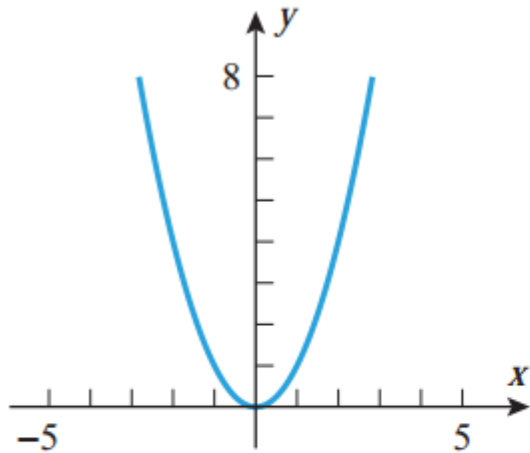
Table 0.2.2

TRANSLATION PRINCIPLES

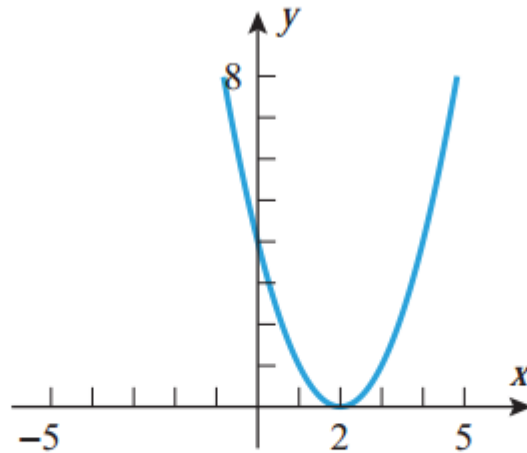
OPERATION ON $y = f(x)$	Add a positive constant c to $f(x)$	Subtract a positive constant c from $f(x)$	Add a positive constant c to x	Subtract a positive constant c from x
NEW EQUATION	$y = f(x) + c$	$y = f(x) - c$	$y = f(x + c)$	$y = f(x - c)$
GEOMETRIC EFFECT	Translates the graph of $y = f(x)$ up c units	Translates the graph of $y = f(x)$ down c units	Translates the graph of $y = f(x)$ left c units	Translates the graph of $y = f(x)$ right c units
EXAMPLE				

Nasrina Parvin

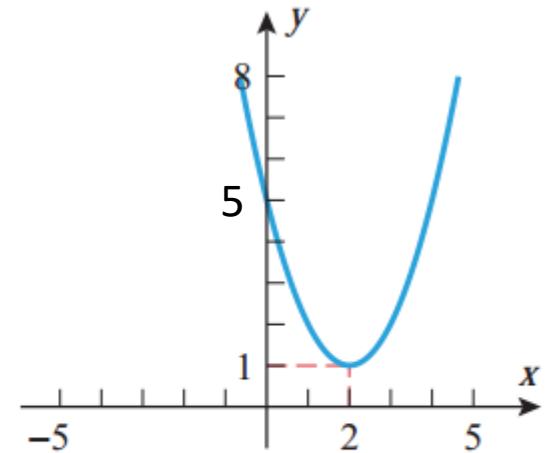
Finding domain and range of $y = f(x) = (x - 2)^2 + 1$



$$y = x^2$$



$$y = (x - 2)^2$$



$$y = (x - 2)^2 + 1$$

Domain is $(-\infty, +\infty)$ or $\mathbb{R}$, and Range is $[1, +\infty)$

Intercepts

- If we put $x = 0$, we get y intercepts, if we put
- $y = 0$, we get x intercepts.

Find the intercepts of $f(x) = x^2 - 4x$. If we put $y = 0$, $y = f(x) = x^2 - 4x = 0$, if $y = 0$

$$x(x - 4) = 0$$

$\Rightarrow x = 0, x = 4$, so x intercepts are $(x, y) = (0, 0)$ and $(4, 0)$, If we put $x = 0$, $y = f(0) = 0^2 - 4 \cdot 0 = 0$, so y intercepts are $(x, y) = (0, 0)$

Nasrina Parvin

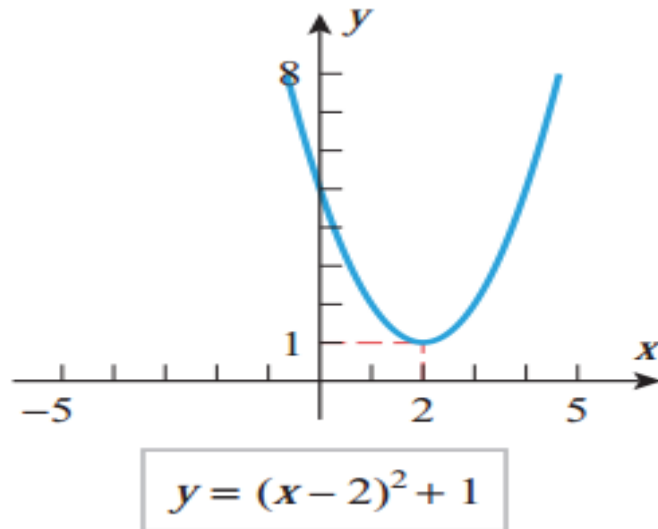
Draw the graph of $f(x) = x^2 - 4x + 5$,

- Example 8:

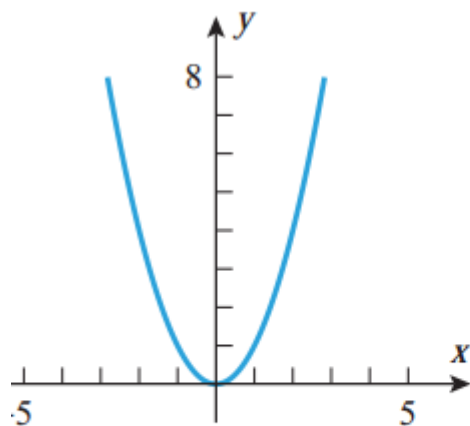
Now,

$$\begin{aligned} f(x) &= x^2 - 4x + 5 = x^2 - 2 \cdot x \cdot 2 + 2^2 + 1 \\ &= (x - 2)^2 + 1 \end{aligned}$$

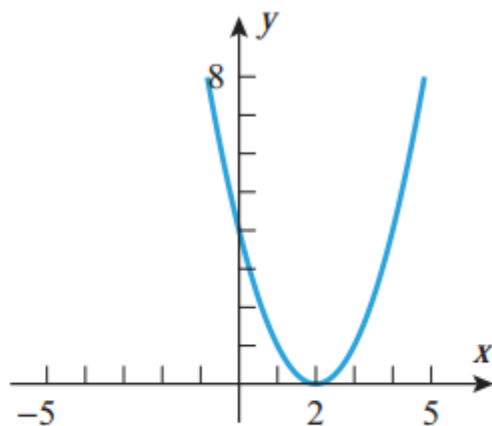
5



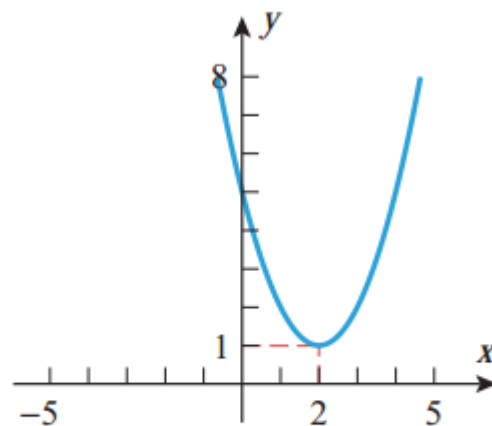
Translations



$$y = x^2$$



$$y = (x - 2)^2$$



$$y = (x - 2)^2 + 1$$

Sketch the graph of $f(x) = x^2 - 4x$, also find their domain and range.

- Now, $f(x) = x^2 - 4x = x^2 - 2 \cdot x \cdot 2 + 2^2 - 4 = (x - 2)^2 - 4$

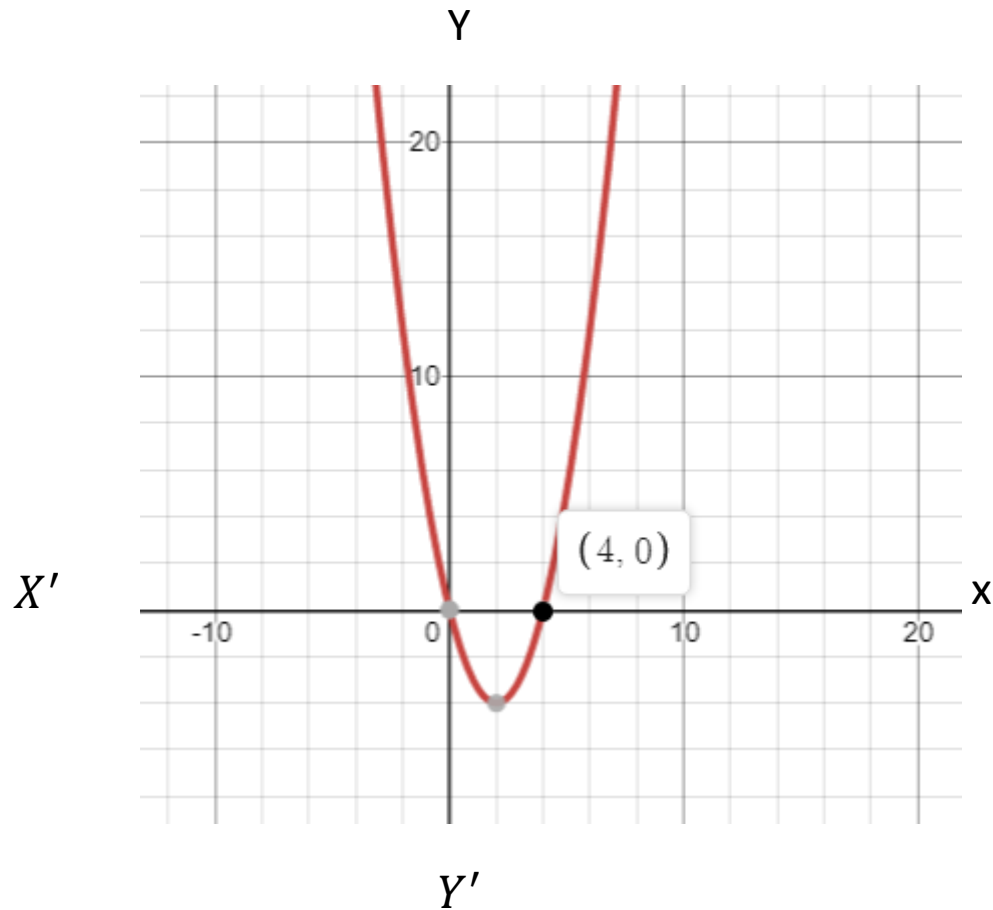
If we put $y = 0$, $y = f(x) = x^2 - 4x = 0$, if $y = 0$
$$x(x - 4) = 0$$

$\Rightarrow x = 0, x = 4$, so x intercepts are $(x, y) = (0, 0)$ and $(4, 0)$, If we put $x = 0$, $y = f(0) = 0^2 - 4 \cdot 0 = 0$, so y intercepts are $(x, y) = (0, 0)$

If we put $y = 0$, $y = f(x) = x^2 - 4x = 0$, if $y = 0$
$$x(x - 4) = 0$$

$\Rightarrow x = 0, x = 4$, so x intercepts are $(x, y) = (0, 0)$ and $(4, 0)$, If we put $x = 0$, $y = f(0) = 0^2 - 4 \cdot 0 = 0$, so y intercepts are $(x, y) = (0, 0)$

Graph of $f(x) = x^2 - 4x$



REF LECTIONS

- [REF LECTIONS\(Page-21\):](#)
- The graph of $y = f(-x)$ is the reflection of the graph of $y = f(x)$ about the y -axis because the point (x, y) on the graph of $f(x)$ is replaced by $(-x, y)$. Similarly, the graph of $y = -f(x)$ is the reflection of the graph of $y = f(x)$ about the x -axis because the point (x, y) on the graph of $f(x)$ is replaced by $(x, -y)$ [the equation $y = -f(x)$ is equivalent to $-y = f(x)$]. This is summarized in Table 0.2.3.

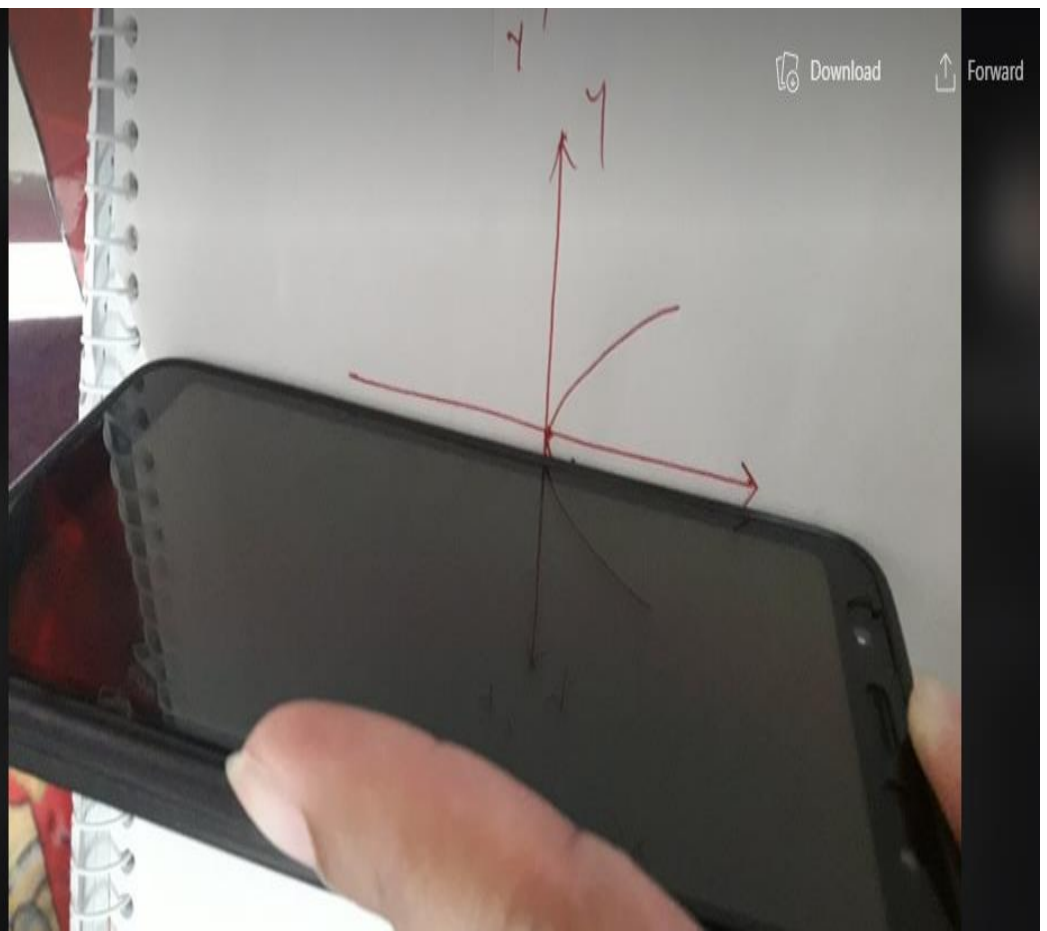
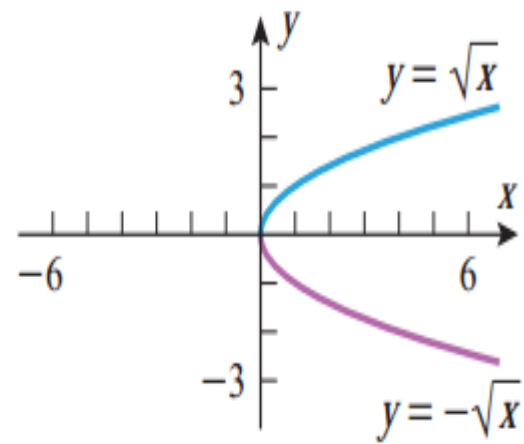
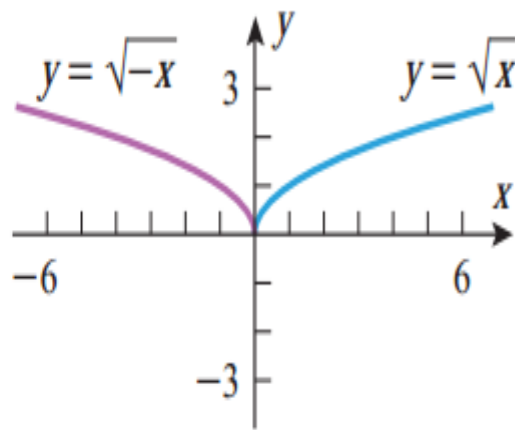
REFLECTIONS

Table 0.2.3

REFLECTION PRINCIPLES

OPERATION ON $y = f(x)$	Replace x by $-x$	Multiply $f(x)$ by -1
NEW EQUATION	$y = f(-x)$	$y = -f(x)$
GEOMETRIC EFFECT	Reflects the graph of $y = f(x)$ about the y -axis	Reflects the graph of $y = f(x)$ about the x -axis

EXAMPLE

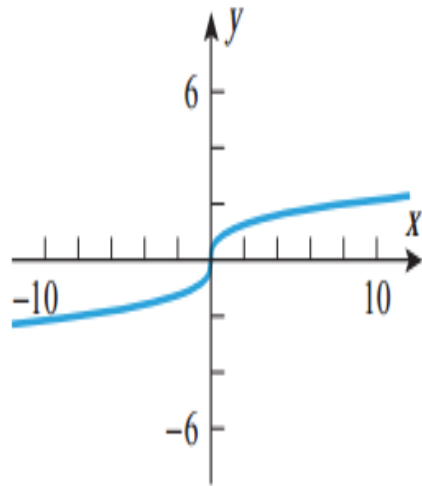


Chapter 0.2

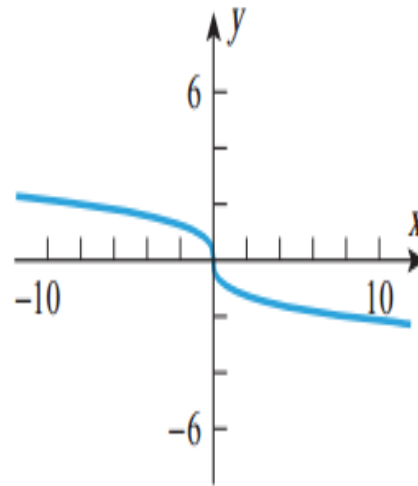
- Example 9: Sketch the graph of $f(x) = \sqrt[3]{2 - x}$.
-

Solution. Using the translation and reflection principles in Tables 0.2.2 and 0.2.3, we can obtain the graph by a reflection followed by a translation as follows: First reflect the graph of $y = \sqrt[3]{x}$ about the y -axis to obtain the graph of $y = \sqrt[3]{-x}$, then translate this graph right 2 units to obtain the graph of the equation $y = \sqrt[3]{-(x - 2)} = \sqrt[3]{2 - x}$ (Figure 0.2.5).

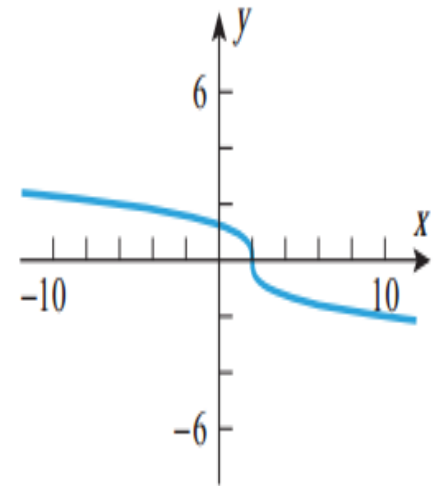
Graph of $f(x) = \sqrt[3]{2-x}$



$$y = \sqrt[3]{x}$$



$$y = \sqrt[3]{-x}$$

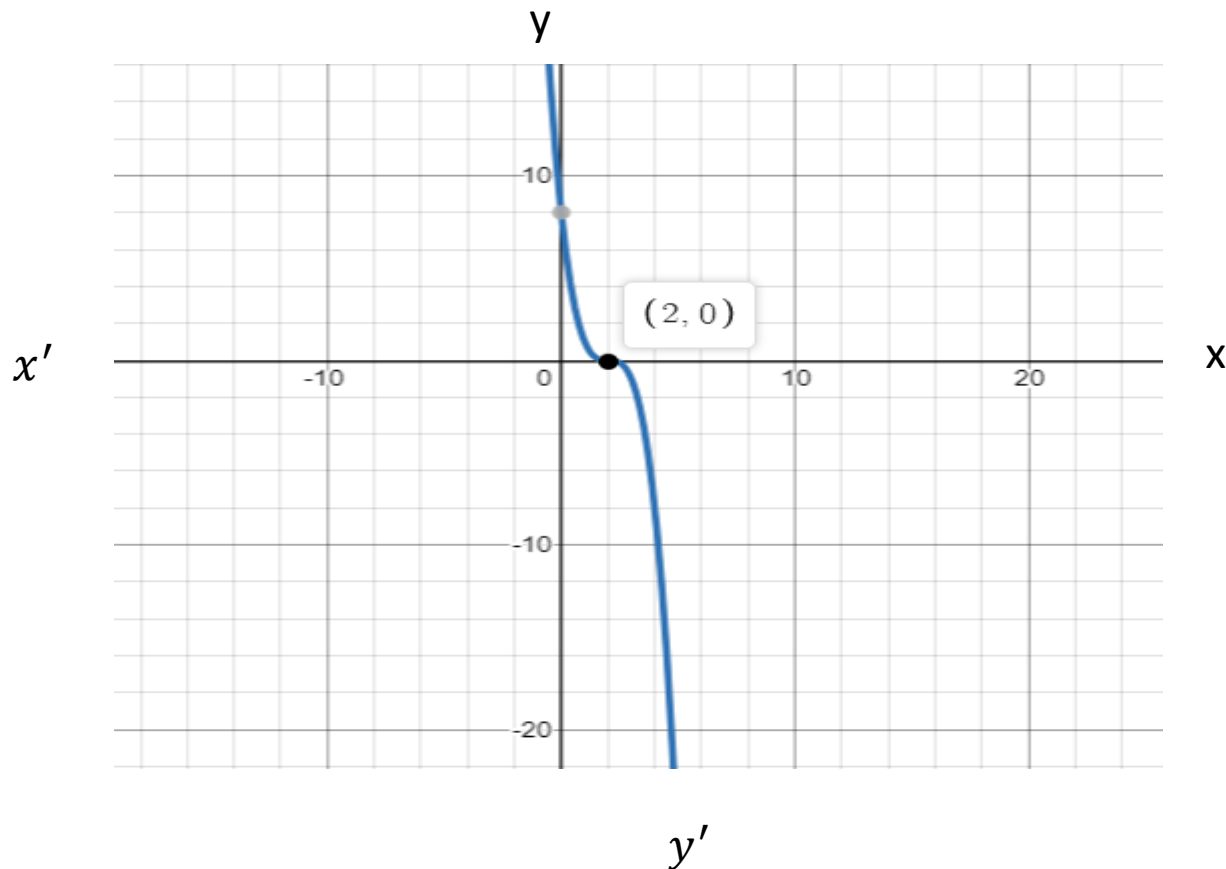


$$y = \sqrt[3]{2-x}$$

► Figure 0.2.5

Translations and reflections

- Sketch the graph of $f(x) = (2 - x)^3$



Exercise:0.1

7. Find $f(0)$, $f(2)$, $f(-2)$, $f(3)$, $f(\sqrt{2})$, and $f(3t)$.

$$(a) \quad f(x) = 3x^2 - 2 \qquad (b) \quad f(x) = \begin{cases} \frac{1}{x}, & x > 3 \\ 2x, & x \leq 3 \end{cases}$$

$$7. (a) \quad f(0) = 3(0)^2 - 2 = -2; f(2) = 3(2)^2 - 2 = 10; f(-2) = 3(-2)^2 - 2 = 10; f(3) = 3(3)^2 - 2 = 25; f(\sqrt{2}) = 3(\sqrt{2})^2 - 2 = 4; f(3t) = 3(3t)^2 - 2 = 27t^2 - 2.$$

$$(b) \quad f(0) = 2(0) = 0; f(2) = 2(2) = 4; f(-2) = 2(-2) = -4; f(3) = 2(3) = 6; f(\sqrt{2}) = 2\sqrt{2}; f(3t) = 1/(3t) \text{ for } t > 1 \text{ and } f(3t) = 6t \text{ for } t \leq 1.$$

Homework

8. Find $g(3)$, $g(-1)$, $g(\pi)$, $g(-1.1)$, and $g(t^2 - 1)$.

$$(a) \ g(x) = \frac{x+1}{x-1} \qquad (b) \ g(x) = \begin{cases} \sqrt{x+1}, & x \geq 1 \\ 3, & x < 1 \end{cases}$$